FREE GLOBAL SECTIONS AFTER AN AFFINE ETALE NEIGHBOURHOOD FOR A FLAT FINITE TYPE MODULE, FINITE PRESENTATION CASE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The local structure statement and the projective neighbourhood statement, with projectivity equal to freeness over a local ring. Same chapter flat.tex, labels proposition-finite-type-flat-at-point (proof line 3491) and proposition-finite-presentation-flat-at-point (proof line 3494); Algebra, label algebra-theorem-projective-free-over-local-ring (proof line 3496). The first reduces to a module of finite presentation on a neighbourhood, the second provides the affine etale neighbourhood with projective global sections, and the third turns projectivity over the local base into freeness.

Source reference. Original theorem 0593: sources/stacks-chow/tags.txt, source lines 6801–6801. Navigation pointer for \ref{algebra-theorem-projective-free-over-local-ring} used inside the counted span (line 3496); the argument that consumes it is retained in the span.

Source reference. Original proposition 05ID: sources/stacks-flat/pinned-original-source.tex, source lines 3287–3287. Navigation pointer for \ref{proposition-finite-presentation-flat-at-point} used inside the counted span (line 3494); the argument that consumes it is retained in the span.

Source reference. Original proposition 05I5: sources/stacks-flat/pinned-original-source.tex, source lines 2477–2477. Navigation pointer for \ref{proposition-finite-type-flat-at-point} used inside the counted span (line 3491); the argument that consumes it is retained in the span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $x \in X$ with image $s \in S$. Assume that*

(1) *f is locally of finite presentation,*
(2) *$\mathcal{F}$ is of finite type, and*
(3) *$\mathcal{F}$ is flat at x over S.*

Then there exists an elementary étale neighbourhood $(S', s') \to (S, s)$ and a commutative diagram of pointed schemes

$$\begin{array}{ccc} (X, x) & \xleftarrow{g} & (X', x') \\ \downarrow & & \downarrow \\ (S, s) & \longleftarrow & (\mathrm{Spec}(\mathcal{O}_{S',s'}), s') \end{array}$$

such that $X' \to X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$ is étale, $\kappa(x) = \kappa(x')$, the scheme X' is affine of finite presentation over $\mathcal{O}_{S',s'}$, the sheaf $g^*\mathcal{F}$ is of finite presentation over $\mathcal{O}_{X'}$, and such that $\Gamma(X', g^*\mathcal{F})$ is a free $\mathcal{O}_{S',s'}$-module.

Proof. To prove the lemma we may replace (S, s) by any elementary étale neighbourhood, and we may also replace S by $\mathrm{Spec}(\mathcal{O}_{S,s})$. Hence by Proposition 05I5 we may assume that $\mathcal{F}$ is finitely presented and flat over S in a neighbourhood of x. In this case the result follows from Proposition 05ID because Algebra, Theorem 0593 assures us that projective = free over a local ring. □